Heston Stochastic Volatility Model

Homework 8 — Simulation, Pricing and Implied Volatility Analysis

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Task 1 — Simulation of the Heston Model

The Heston model is described by the following system of stochastic differential equations:

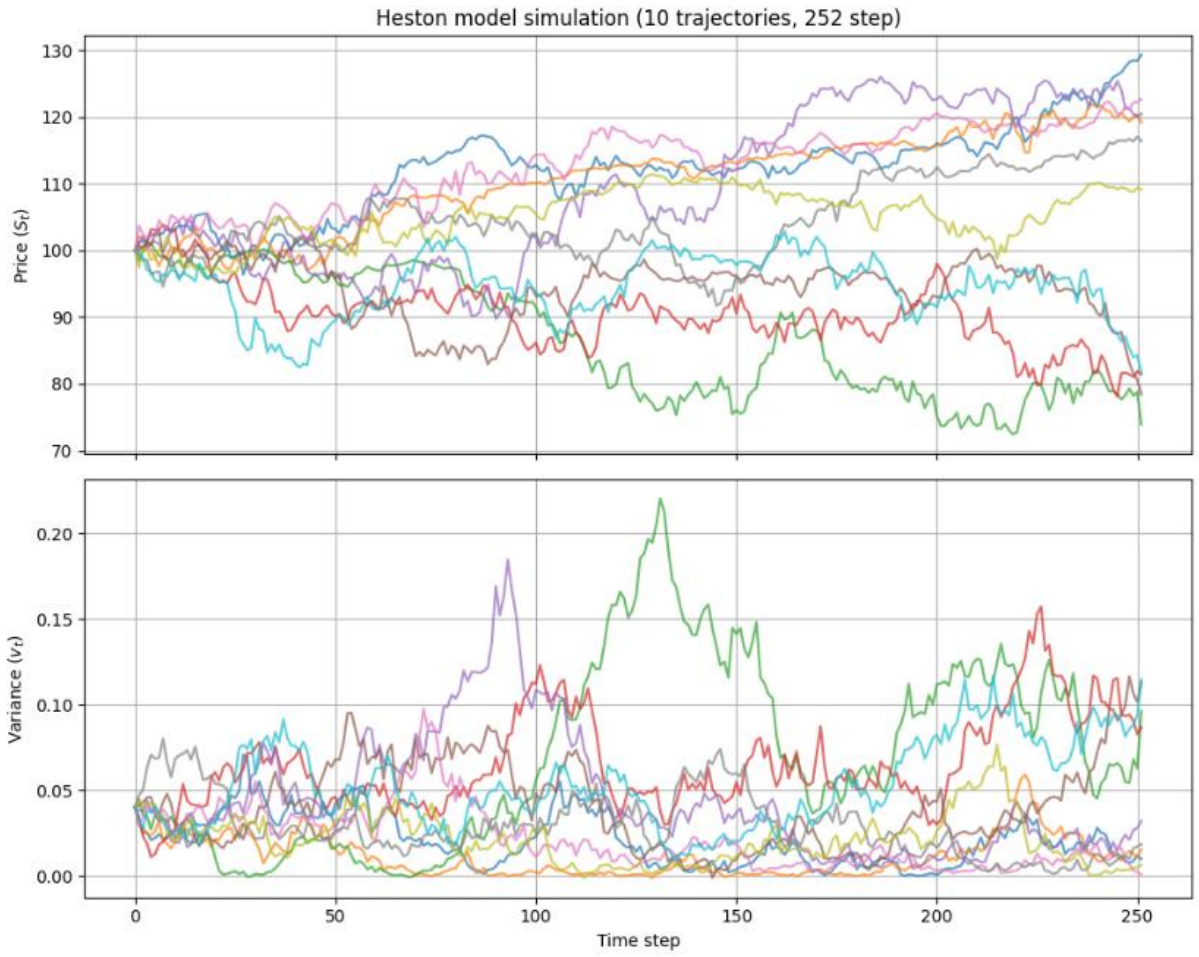
$dS_t = \mu S_t\,dt + \sqrt{v_t} S_t\,dW_t^S \qquad dv_t = \kappa(\theta - v_t)\,dt + \xi\sqrt{v_t}\,dW_t^v$

where the two Brownian motions are correlated with $d\langle W^S, W^v\rangle = \rho\,dt$. The parameters used throughout this homework are:

Parameter	Symbol	Value	Description
Mean reversion speed	κ	2.0	Fast reversion
Long-run variance	θ	0.04	20% long-run vol
Vol of vol	ξ	0.5	High volatility clustering
Correlation	ρ	−0.7	Negative (leverage effect)
Initial variance	v0	0.04	At long-run level

The simulation uses the Euler-Maruyama discretization scheme over N = 250 time steps and M = 10,000 paths with T = 1 year, S0 = 100, μ = 0.02. Correlated Brownian increments are generated via a bivariate normal with covariance matrix [[1, ρ], [ρ, 1]]. To preserve numerical stability, a reflection scheme is applied to the variance process: v_prev = |vt−1| before each Euler step, preventing the square root from becoming undefined when vt turns negative due to discretization error.

The 10 simulated price paths show the typical behaviour of the Heston model: paths start at S0 = 100 and fan out progressively, with some trajectories reaching above 130 and others falling below 70 after one year. The spread is noticeably asymmetric, with downside paths tending to be steeper than upside paths. This asymmetry is a direct consequence of ρ = −0.7: when the price falls, volatility rises (leverage effect), amplifying losses and producing heavier left tails. The variance paths begin at v0 = 0.04 and oscillate around the long-run mean θ = 0.04, with mean reversion governed by κ = 2.0. The high vol-of-vol (ξ = 0.5) causes large transient excursions, with some paths briefly exceeding 0.15–0.20 before being pulled back toward θ.

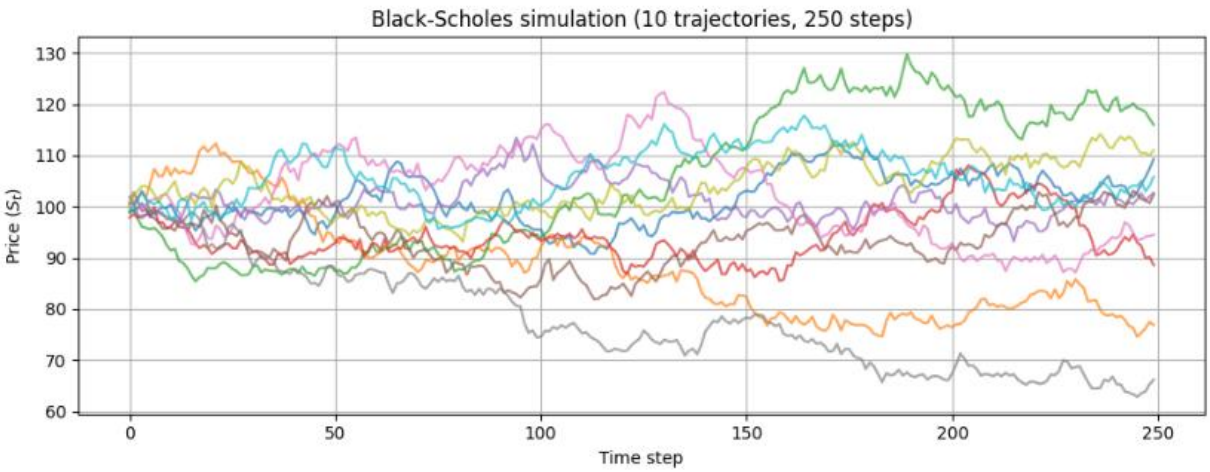


Task 2 — Comparison of Terminal Distributions: Heston vs Black-Scholes

Black-Scholes is simulated with constant volatility σ = √θ = 20%, matching the long-run volatility of the Heston model. The log-normal exact scheme is used for Black-Scholes to avoid Euler discretization bias. Terminal prices ST are collected from M = 10,000 paths for both models and their distributional moments compared:

Statistic	Black-Scholes	Heston
Mean	101.84	102.00
Variance	417.07	355.10
Skewness	0.637	−0.621
Kurtosis	0.875	0.792

Both means are close to the theoretical value S0 · eμT ≈ 102.02, confirming correct simulation. The most striking difference is in skewness: Black-Scholes produces positive skewness (right tail, consistent with a log-normal), while Heston produces negative skewness. This sign reversal is entirely due to ρ = −0.7. Under Heston, price declines are systematically accompanied by volatility spikes, making large losses more likely and large gains less so — generating a left-skewed distribution. The lower variance under Heston (355 vs 417) is also notable: stochastic volatility with strong mean reversion keeps variance from drifting as far as a constant-sigma model allows, effectively compressing the terminal distribution. The kurtosis values are similar, both indicating mild excess kurtosis relative to the Gaussian.



Task 3 — European Call Pricing: Heston vs Black-Scholes

European calls are priced using Monte Carlo under both models. Heston prices are computed as $e^{-rT} \cdot E[\max(S_T - K, 0)]$ using the 10,000 simulated terminal prices. Black-Scholes prices are obtained via the closed-form formula with $\sigma = \sqrt{\theta} = 20\%$.

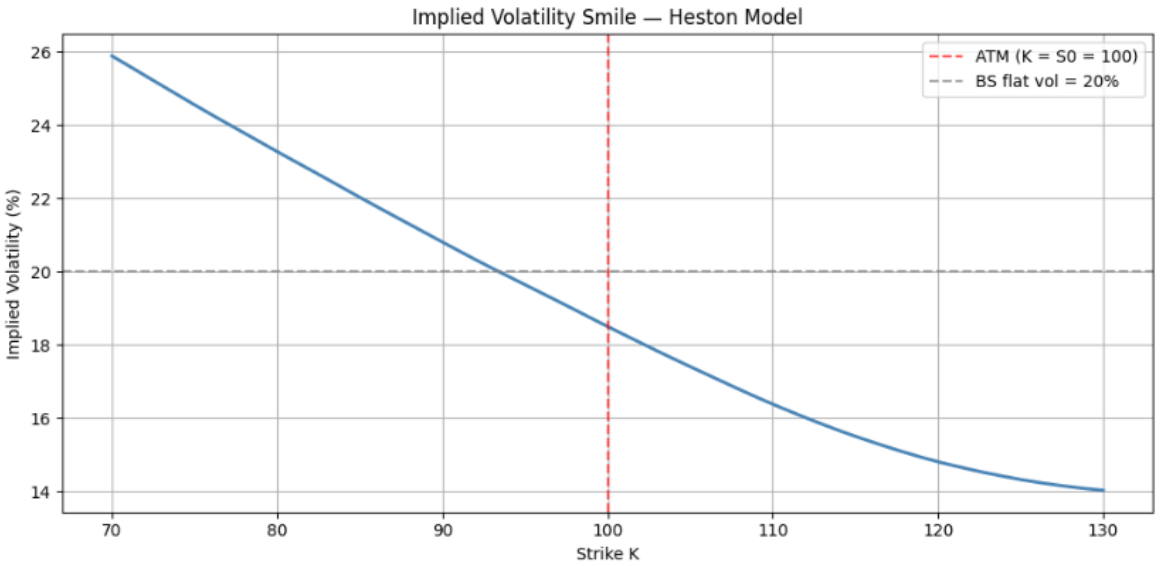
Strike K	BS Price	Heston Price	Difference
90	14.81	15.05	+0.25
100	8.92	8.33	−0.59
110	4.94	3.57	−1.37

The pricing differences reveal a clear pattern driven by the volatility skew. For the in-the-money call (K = 90), Heston prices slightly above Black-Scholes: the negative correlation concentrates probability mass on the left tail, making deep ITM options slightly more valuable due to the higher implied volatility at low strikes. For ATM (K = 100) and OTM calls (K = 110), Heston prices below Black-Scholes. This is because $\rho = -0.7$ reduces the implied volatility at high strikes (OTM calls), making upside scenarios less likely than a flat-vol model assumes. The monotonic decrease of the Heston−BS difference from positive to increasingly negative as K rises is the direct fingerprint of a negative volatility skew, which Black-Scholes — with its flat volatility surface — cannot capture.

Task 4 — Implied Volatility Smile

For each strike K in the grid [70, 130] (50 points), the Heston Monte Carlo call price is computed and then numerically inverted via the Brent method (scipy.optimize.brentq) to find the Black-Scholes implied volatility that would reproduce the same price.

The resulting IV curve is strongly downward sloping (a volatility skew), falling from approximately 26% at K = 70 to around 14% at K = 130, crossing the flat BS reference level of 20% near the ATM point. This shape is the hallmark signature of $\rho < 0$. Intuitively, a negative spot-vol correlation means that when the stock price falls, volatility rises — so low-strike puts (and equivalent calls via put-call parity) become more expensive, inflating their implied volatility. Conversely, high-strike OTM calls are cheaper, giving lower IV. The curve does not form a U-shaped smile (as would be produced by a symmetric $\rho = 0$ case), but instead a monotone skew. This is consistent with observed equity option markets, where the skew is almost universally negative.



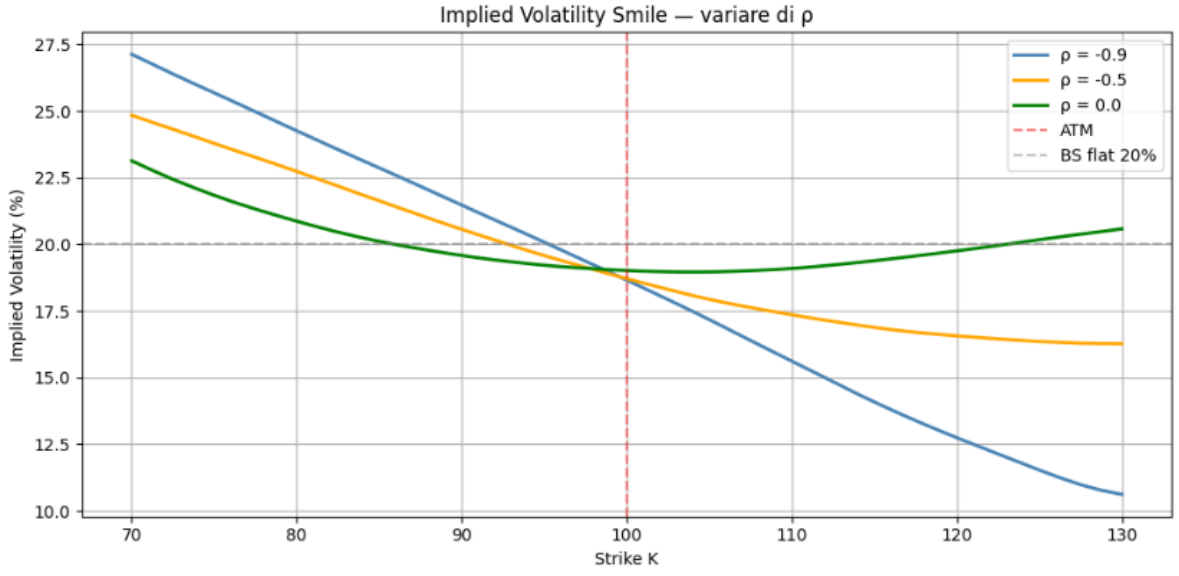
Task 5 — Sensitivity of the Smile to ρ and ξ

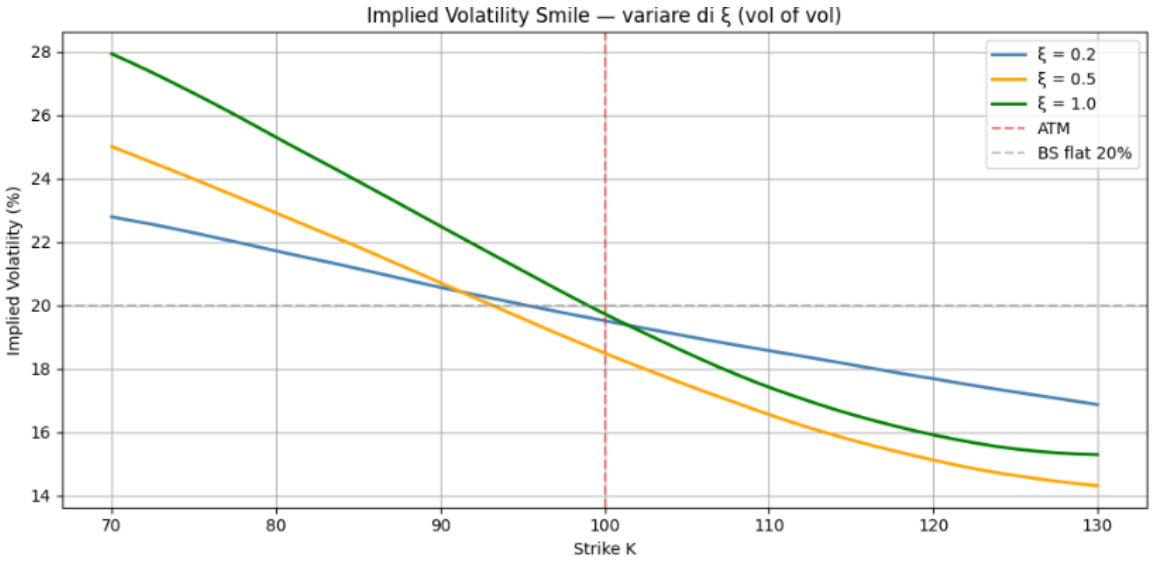
Effect of correlation ρ ∈ {−0.9, −0.5, 0.0}

As ρ becomes more negative, the skew steepens dramatically. With ρ = −0.9, the IV curve drops from ~27.5% at K = 70 to below 11% at K = 130, with the slope much steeper than for ρ = −0.5. With ρ = 0, the curve takes on a mild symmetric U-shape (a true smile) centred at ATM, reflecting the symmetric impact of stochastic volatility with no directional correlation. This confirms that ρ controls the tilt of the smile: negative ρ tilts it into a skew (left arm up, right arm down), while ρ = 0 produces symmetry. All curves roughly intersect near ATM, since the ATM level is primarily set by the total variance budget.

Effect of vol-of-vol ξ ∈ {0.2, 0.5, 1.0}

Higher ξ amplifies the overall curvature and steepness of the skew without changing its directional tilt (which is governed by ρ). With ξ = 0.2, the curve is nearly flat, close to the 20% BS reference — reflecting that low vol-of-vol means variance barely moves, approaching the Black-Scholes world. With ξ = 1.0, the curve is steep and highly asymmetric, with IV reaching ~28% at low strikes. This is because higher vol-of-vol creates fatter tails in both directions, but the interaction with ρ = −0.7 amplifies the left tail more than the right, increasing OTM put prices substantially. In practice, ξ is the primary calibration parameter for the curvature of the smile surface.





Task 6 — Feller Condition and Variance Discretization

The Feller condition for the CIR process (variance under Heston) states that $2\kappa\theta \geq \xi^2$ is sufficient to guarantee that the continuous-time process vt remains strictly positive. With the given parameters:

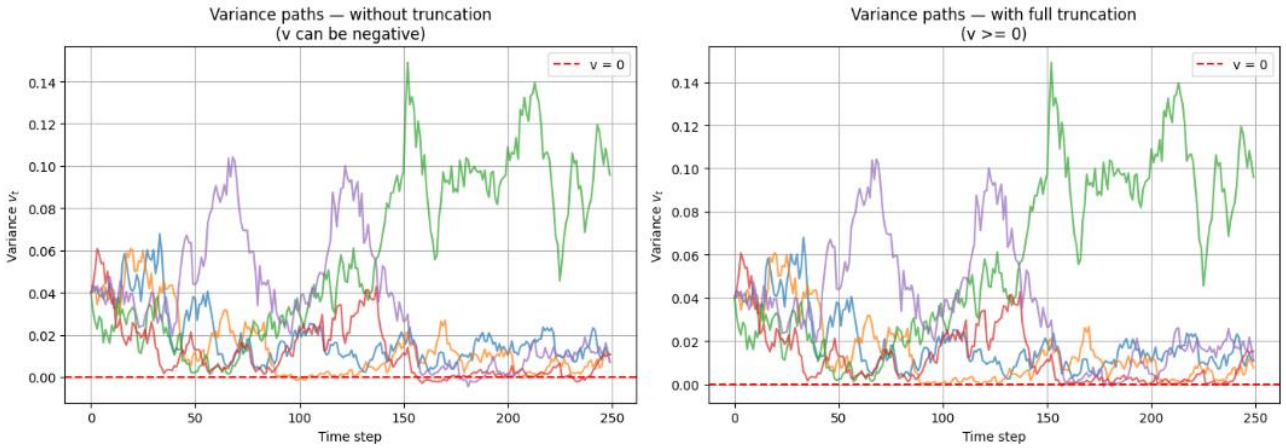
$2 \cdot 2.0 \cdot 0.04 = 0.16 \quad < \quad 0.25 = 0.5^2 \quad \Longrightarrow$ Feller condition VIOLATED

In continuous time, this means zero is attainable, though not necessarily reached with probability 1. In the Euler discretization, the situation is worse: even if Feller were satisfied in continuous time, the finite time-step approximation can produce negative variance values. The simulation (without truncation) shows approximately 2.38% of all variance values across paths and time steps are negative.

Three schemes are compared:

- No truncation (Euler pure): v can become negative. The sqrt(|v|) reflection used in HeMC prevents NaN but does not prevent negative v. This scheme is used to measure the frequency of negativity.
- Full truncation: v_prev = max(v_{t-1}, 0) before each Euler step. This sets negative variances to zero and is the simplest fix. It introduces a mild upward bias in the level of variance near zero but is stable and widely used in practice.
- Reflection (used in main HeMC): v_prev = |v_{t-1}|, which treats negative variance symmetrically by folding it back to positive. This is an alternative that also preserves positivity.

The plots confirm that without truncation, paths frequently dip below zero (red dashed line), especially during volatile periods. With full truncation, all paths stay non-negative, though they can be seen ‘stuck’ at zero for brief periods during which the drift (mean reversion) works to push them back up. The lesson is clear: Euler-Maruyama is not positivity-preserving for the CIR process and corrections are always needed in practice. More sophisticated schemes (e.g. Milstein, QE scheme of Andersen) exist but are beyond the scope of this homework.



Task 7 — Fourier Pricing via the Characteristic Function

Characteristic function

The Heston model admits a closed-form characteristic function for the log-price log(ST), which makes semi-analytical Fourier pricing possible. The characteristic function φ(u) = E[e^{iu log(ST)}] has the Heston (1993) form:

φ(u) = exp(C(u) + D(u)·v0 + iu·log(S0))

where C and D are known functions of the model parameters involving complex exponentials. The numerically stable form (avoiding branch-cut issues) keeps the sign convention γ = κ − ρξ iu − d in the numerator and denominator of g, ensuring continuity of log(1 − g e^{−dT}).

Fourier pricing: Lewis (2001) formula

The European call price is computed via:

C = S0 − (K e^{−rT} / π) · ∫₀^∞ Re[e^{+iuk} φ(u − i/2) / (u² + 1/4)] du

where k = log(K e^{−rT} / S0) is the log-moneyness. The integral is approximated with the trapezoidal rule on u ∈ [10^{-5}, 100] with 5,000 points. A critical detail: the sign of the exponential must be +iuk (not −iuk), otherwise the integral diverges for OTM calls and produces negative prices — a common implementation mistake.

Results: Fourier vs Monte Carlo

Strike K	C_Fourier	C_MC (100k paths)	Abs Error
80	23.1882	23.1142	0.0741 (< 0.10)
90	15.0921	15.0476	0.0446 (< 0.10)
100	8.3358	8.3466	0.0108 (< 0.10)
110	3.5706	3.6353	0.0647 (< 0.10)
120	1.1068	1.1669	0.0601 (< 0.10)

The Fourier method is substantially more accurate than Monte Carlo for vanilla European options on Heston. Its error is purely from the discretization of the integration grid (controllable by increasing N_int), while Monte Carlo error is statistical of order O(1/√M), so approximately ±0.05–0.10 with M = 100,000 paths. Crucially, Fourier is also orders of magnitude faster: a single call takes microseconds, versus seconds for large Monte Carlo simulations. The trade-off is that Fourier pricing requires a closed-form characteristic function — which exists for Heston — but is not applicable to path-dependent options (barrier, Asian, lookback) where Monte Carlo remains the only general approach.

References:

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